

AETHER BIONICS

AI Usage Declaration & Technical Overview

Landing Page Deliverables - Web Development Internship Submission

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Project Overview

Aether Bionics is a landing page designed and developed for a deep-tech bionics firm specializing in two distinct but technologically convergent domains: **Humanoid Robotics** (Aegis Platform) and **Rehabilitation Devices** (NeuroPulse Platform). The landing page achieves symmetric visual weight, presenting both domains as equal pillars of a unified engineering ecosystem.

- **Stack:** React (v18) + Vite (v5) + CSS3 (Vanilla)
- **Interactive Components:** Canvas Neural Net Background, Split-Screen Resizing Hero, Symmetrical Feature Grid, Specification Comparator Tabs, and a Dynamic Inquiry Form with responsive inputs.
- **Visual Design:** High-fidelity dark mode with neon glow accent states, glassmorphic overlays, custom scrollbar styling, and smooth scroll-triggered entry animations.

Detailed AI Usage Declaration

In accordance with the submission guidelines, this document outlines how and where Antigravity AI was utilized during the design and development phases.

1. Architectural Design & Scaffolding

- **Scaffolding:** The AI was used to check `create-vite` CLI options (`npx create-vite@latest --help`) and script the non-interactive React project initialization.
- **Component Design Pattern:** The AI architected a modular React structure, dividing the single-page application into dedicated interactive layout modules (`SplitHero`, `ParticleBackground`, `TechShowcase`, `SpecComparison`, `SynergySection`, `ContactForm`).

2. High-Fidelity UI/UX & Styling

- **Interactive Split-Screen:** The AI authored the custom CSS and React state tracking that enables the 50/50 hero layout. Hovering over a domain expands it to 65% width while scaling down the other, shifting a glowing bionic split divider across the screen.
- **Glassmorphism & Neon Shadows:** The AI designed the custom theme variables in ``src/index.css``, employing blur backdrops (``backdrop-filter``) and glow shadow transitions (``box-shadow: 0 0 25px rgba(0, 240, 255, 0.35)``).
- **Responsive Systems:** The AI constructed CSS media query break-points converting the horizontal split layout into a mobile vertical stack, maintaining symmetrical visual weight across both columns on smartphones.

3. Media Asset Generation (AI Artistry)

To avoid generic template placeholders, the AI's image generation tools were leveraged to build custom high-quality bionic assets:

- ``robot.png``: A photorealistic, high-resolution render of a humanoid bionic android with soft cyan joint lights.
- ``rehab.png``: A photorealistic, high-resolution render of an advanced wearable hand/forearm bionic exoskeleton sleeve with soft violet joint lights.
- The prompt structures, lighting definitions (cinematic, 8k), and aesthetics were crafted to align with the deep-tech brand.

4. Interactive Logic & Algorithms

- **Neural Net Background:** The AI coded a canvas-based node connection script in React. Particles float inside the viewport and draw glowing vector connectors when near each other, dynamically drawing a connection path to the user's cursor as it moves.
- **Scroll Observer:** The AI wrote a React-based ``IntersectionObserver`` hook in ``src/App.jsx`` that automatically flags elements with ``scroll-anim`` as ``animated`` when they scroll into the viewport, triggering smooth fade-and-rise transitions.
- **Inquiry Form Validation:** The AI wrote the state-driven validation script and responsive feedback alerts for the contact panel, including a simulated network delay and custom success state overlay.

5. Submission Packaging

- **PDF Compilation:** The AI engineered a Python script utilizing ``fpdf2`` to read this markdown file and compile it into a clean, formatted PDF (``ai_usage.pdf``) for upload.

Technical Features Summary

Metric / Feature	Humanoid Robotics (Aegis Platform)	Rehabilitation Devices (NeuroPulse)
Primary Theme Color	Cyber Cyan (<code>`#00f0ff`</code>)	Radiant Violet/Magenta (<code>`#d946ef`</code>)
Degrees of Freedom	44-DoF Actuated Joints	Up to 12-DoF Active Assist



****Telemetry Latency****



< 2.5ms

< 1.2ms (Surface EMG translation)



****Compliance Standard****



Industrial Safety Envelope

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FDA Class II Medical Compliance

Google Drive Submission Links:

Please refer to the candidate's actual Google Drive submission links. The codebase is organized within the local workspace under '/scratch/bionics-landing-page', containing the complete production React + Vite components and custom CSS styling.